NVC

Quick Start Guide

Version 2.3.0

Language: en

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Before You Start

NVC converts video using FFmpeg. It needs two things before it can do anything: a source folder to watch, and a destination folder to write to.

Note

- ffmpeg.exe and ffprobe.exe must sit alongside the application files. NVC reports them as missing at startup if they are not there.

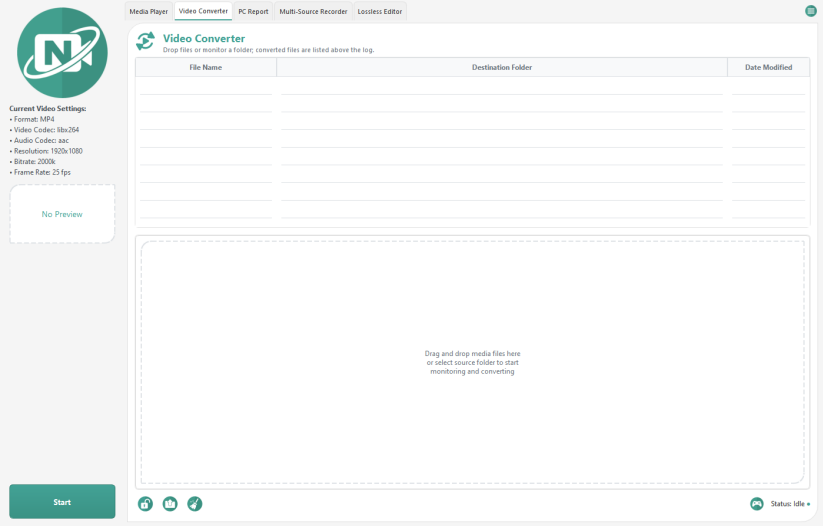


Figure 0. The NVC main window, with the conversion log and preview panel.

Your First Conversion

1. Open Settings from the menu.
2. Set a Source Folder -- the folder NVC watches for new video.
3. Set a Destination Folder -- where converted video is written.
4. Leave the output format at MP4 and the codecs at their defaults. These suit almost every case.
5. Close Settings and click Start.
6. Copy a video into the source folder. NVC queues it and converts it automatically.

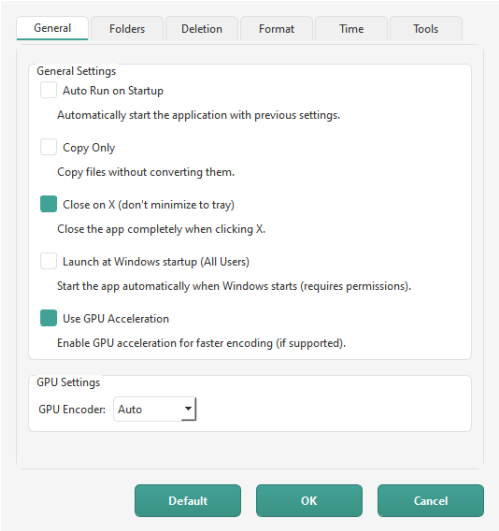


Figure 1. Settings, General tab: startup, Copy Only, tray and GPU options.

Progress appears in the conversion log. When the job finishes, the Converted Files tab records the result.

Record From a Webcam

The Multi-Source Recorder captures up to four cameras or streams. One USB webcam is the simplest case.

1. Open the Multi-Source Recorder tab and click Video Settings.
2. In row 1, open the Video device list and pick your camera by its Windows name. If it is missing, choose Refresh USB devices...
3. Pick a microphone in Audio device, or No Audio. Leave Frame rate and Resolution at the detected values. If detection failed, leave Resolution at Auto and choose a frame rate the camera supports.
4. Click OK. Back on the tab, choose Separate videos and a File location.
5. Leave Timer blank, then click Start Recording in the footer. Click Stop Recording when you are done.

The file is written to the File location as Video_<Video name>_<timestamp>.mp4. In Video Settings, use Save as... to save the setup as a preset so the next session is one Load away.

Review a Recording

1. In Converted Files, select a file and click Play, or right-click it and choose Play in Media Player. You can also open the Media Player tab and click Load on a panel.
2. Press Space to play or pause, Left and Right to step one frame, Shift+Left and Shift+Right to jump 5 seconds.
3. Press F for fullscreen and Escape to come back.
4. Set Panels to 1, 2 or 4. Use 2 or 4 to compare recordings side by side; each panel has its own offset to line up clips that started at different times.

Note

- The Media Player tab is hidden by default. Play in Media Player shows it for this session; tick it in Settings, Tools tab, to keep it.

Settings Worth Knowing

Setting	Start with
Output Format	MP4. The most widely playable container.
Video Codec	libx264. Plays essentially everywhere.
Audio Codec	AAC. The standard partner for MP4.
Video Bitrate	Auto, unless you are targeting a specific file size.
Video Size	Auto, to keep the original dimensions.
Original Files	Keep Originals, until you trust the output.

Note

- NVC never overwrites an existing output. A second conversion of video.mp4 becomes video (1).mp4, exactly like a repeated browser download.

If Something Goes Wrong

1. Read the conversion log first. FFmpeg usually says exactly what it objected to.
2. Check the app is unlocked and monitoring is started.
3. Confirm the file extension is one NVC supports.
4. If GPU encoding is on, turn it off and retry on the CPU.
5. For a failure with no message at all, set the NVC_LOG_LEVEL environment variable to DEBUG and reproduce it. NVC then logs the full traceback.

The Help manual covers every tab and setting in full, including the Lossless Editor, the Multi-Source Recorder, and the PC Report tool.

Glossary

Terms used throughout this guide. FFmpeg terminology is kept where it is what you will see in the conversion log.

Term	Meaning
Container	The file wrapper that holds video and audio streams, such as MP4, MKV or AVI. The container is what the file extension names; it is separate from the codec used inside it.
Codec	The scheme used to compress a stream. H.264 (libx264) and HEVC (libx265) are video codecs; AAC and MP3 are audio codecs.
FFmpeg	The command-line media tool NVC runs to do the actual conversion. Its output is what appears in the conversion log.
FFprobe	FFmpeg's inspection tool. NVC uses it to read duration, streams, codecs and other metadata without converting anything.
Stream copy	Rewriting media into a new container without re-encoding it. Fast and lossless, but only possible when the target container supports the existing streams. Also called remuxing.
Remux	Changing a file's container while keeping its streams untouched. See stream copy.
Transcode	Decoding and re-encoding a stream, which changes quality and takes significantly longer than a stream copy.
Keyframe	A frame stored in full rather than as a difference from its neighbours. Seeking and lossless trimming can only land cleanly on a keyframe.
GOP	Group of Pictures: the interval between keyframes. A shorter GOP makes scrubbing more precise at the cost of file size.
Bitrate	Data used per second of media, usually in kbps or Mbps. Higher bitrate generally means better quality and a larger file.
GPU encoding	Using dedicated video hardware (NVENC on NVIDIA, AMF on AMD, Quick Sync on Intel) to encode instead of the CPU. Much faster, with codec support depending on your drivers.
DirectShow	The Windows capture interface NVC uses to record from local devices. Sources are written as video=Device Name.
Media Foundation	The Windows playback framework the Media Player uses to decode and play supported media.
RTSP	A streaming protocol used by most IP and network cameras. Sources are written as a URL, such as rtsp://192.168.1.10/stream.
GenICam	The generic camera interface used by Basler and other industrial cameras. Configure these cameras in Basler pylon Viewer.
Preset	A saved recorder configuration stored as JSON under Public\Documents\Noldus\NVC\MediaRecorder\Presets.
Sync offset	A time adjustment that shifts one source earlier or later in a Combined grid or Picture-in-picture recording so sources line up.

Picture-in-picture	A recorder output mode: source 1 fills the frame and the other enabled sources appear as small insets along the bottom left, in one file.
Screen capture	A recorder source type that records the whole desktop, one display, one window or a region of the screen instead of a camera.
Frame step	Moving exactly one frame forward or backward using the clip's frame rate.
Playlist	An ordered list of media files for one Media Player panel. With one panel, the next item starts automatically.
Administrator (elevation)	Running NVC with administrator rights so the PC report can read protected Windows data.
Tool tab	A main-window tab that can be hidden or shown in Settings -- Tools.
Retention period	How long NVC keeps an original file after a successful conversion before deleting it. Pending deletions survive a restart.
Redaction	Always-on masking of sensitive values such as licence keys, serial numbers, MAC addresses and user paths in a PC report before it is shared.